Introduction to (Statistical) Machine Learning
Problem Set Two

Summer 2026, IIM B

Due by 11:59 pm on 6 July 2026

Instructions. All problems are required. For theoretical problems, using LLM's is not allowed. You are free to use them for the coding problems.

1. Consider the multiple regression model:

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_m x_{im} + \epsilon_i \qquad \text{with } \epsilon_i \overset{\text{i.i.d}}{\sim} N(0, \sigma^2). \tag{1}$$

We observe data $(x_{i1}, \dots, x_{im}, y_i), i = 1, \dots, n$. In the model above, the covariates x_{ij} are treated as fixed constants. For Bayesian analysis with the prior,

$$\beta_0, \beta_1, \dots, \beta_m, \log \sigma \overset{\text{i.i.d}}{\sim} \text{Unif}(-C, C).$$

we showed (in Lecture 15) that when $C \to \infty$,

$$\beta \mid \text{data} \sim t_{n-m-1,m+1}\left(\hat{\beta}, \hat{\sigma}^2 (X^T X)^{-1}\right) \tag{2}$$

where $X, \beta, \hat{\beta}, \hat{\sigma}$ are all as defined in Lecture 15.

This exercise provides another slightly different proof of this same result. Along the way, it also discusses inference for the parameter σ. Throughout, assume that $C \to \infty$ (i.e., C is very large).

a) Prove that

$$\beta \mid \text{data}, \sigma \sim N_{m+1}\left(\hat{\beta}, \sigma^2 \left(X^T X\right)^{-1}\right). \tag{3}$$

The left hand side above is referring to the conditional density of β given the data as well as the parameter σ.

b) Prove that the posterior density of σ (i.e., the density of $\sigma \mid$ data) is proportional to

$$\sigma^{-n+m} \exp\left(-\frac{S(\hat{\beta})}{2\sigma^2}\right) I\{\sigma > 0\}, \tag{4}$$

where $S(\hat{\beta})$ is the Residual Sum of Squares.

c) Use the results of the previous two parts along with the formula

$$f_{\beta|\text{data}}(\beta) = \int_0^\infty f_{\beta|\text{data},\sigma}(\beta) f_{\sigma|\text{data}}(\sigma) d\sigma$$

to deduce (2).

d) Use the result in part (b) to show that

$$\frac{S(\hat{\beta})}{\sigma^2} \mid \text{data} \sim \chi^2_{n-m-1}.$$

e) Use the result in the previous part to argue that the usual estimator:

$$\hat{\sigma} := \sqrt{\frac{S(\hat{\beta})}{n-m-1}}$$

is a reasonable point estimate of σ.

2. This problem discusses predictions in linear regression. Consider the same setting as the previous problem.

a) Suppose we are interested in the parameter $a^T\beta$ for some fixed vector a. Using (3), prove that

$$a^T\beta \mid \text{data}, \sigma \sim N\left(a^T\hat{\beta}, \sigma^2 a^T \left(X^TX\right)^{-1} a\right)$$

Note that we are conditioning on the data as well as the parameter σ in the left hand side above.

b) Using (2), prove that

$$a^T\beta \mid \text{data} \sim t_{n-m-1}\left(a^T\hat{\beta}, \hat{\sigma}^2 a^T \left(X^TX\right)^{-1} a\right)$$

c) Suppose we are interested in predicting the value Y_{n+1} of the response variable at a new set of covariate observations $x_{n+1,1}, \dots, x_{n+1,m}$. Assuming that

$$Y_{n+1} = \beta_0 + \beta_1 x_{n+1,1} + \cdots + \beta_m x_{n+1,m} + \epsilon_{n+1}$$

for an independent error $\epsilon_{n+1} \sim N(0, \sigma^2)$, prove that

$$Y_{n+1} \mid \text{data}, \sigma \sim N\left(\tilde{x}_{n+1}^T\hat{\beta}, \sigma^2 + \sigma^2\tilde{x}_{n+1}^T \left(X^TX\right)^{-1} \tilde{x}_{n+1}\right)$$

where $\tilde{x}_{n+1}$ is the vector with entries $1, x_{n+1,1}, \dots, x_{n+1,m}$.

d) Using the result from the previous part, the formula

$$f_{Y_{n+1}|\text{data}}(y) = \int_0^\infty f_{Y_{n+1}|\text{data},\sigma}(y) f_{\sigma|\text{data}}(\sigma) d\sigma,$$

and the density $\sigma \mid$ data from (4), prove that

$$Y_{n+1} \mid \text{data} \sim t_{n-m-1}\left(\tilde{x}_{n+1}^T\hat{\beta}, \hat{\sigma}^2 + \hat{\sigma}^2\tilde{x}_{n+1}^T \left(X^TX\right)^{-1} \tilde{x}_{n+1}\right).$$

This result can be used for obtaining predictions of Y_{n+1} along with uncertainty intervals.

3. Consider the `spam7.csv` dataset which consists of data on $n = 4601$ emails corresponding to the following variables:

- `crl.tot`: Total length of uninterrupted sequences of capital letters.

- `dollar`: Occurrences of the dollar sign ($), expressed as a percentage of the total number of characters.
- `bang`: Occurrences of the symbol `!`, expressed as a percentage of the total number of characters.
- `money`: Occurrences of the word "money", expressed as a percentage of the total number of words.
- `n000`: Occurrences of the string "000", expressed as a percentage of the total number of words.
- `make`: Occurrences of the word "make", expressed as a percentage of the total number of words.
- `yesno`: Indicates whether the message is spam, with levels

$$n = \text{not spam}, \qquad y = \text{spam}.$$

Our goal is to perform logistic regression on this dataset without using any existing libraries. Our response variable y will be the `yesno` variablle which indicates whether the email is spam or not. As covariates, use the following five variables

$$x_1 = \log(\texttt{crl.tot}) \quad x_2 = \log(\texttt{dollar} + 0.001) \quad x_3 = \log(\texttt{bang} + 0.001)$$
$$x_4 = \log(\texttt{money} + 0.001) \quad x_5 = \log(\texttt{n000} + 0.001) \quad x_6 = \log(\texttt{make} + 0.001)$$

We are taking logarithms because these variables are highly skewed otherwise. Also we are adding 0.001 because some of these variables can take the value 0 and we don't want $\log(0)$.

a) Using the Newton's algorithm, calculate the maximum likelihood estimate of the coefficients in logistic regression. Check the correctness of your solution using an inbuilt library function for fitting logistic regression.

b) Compute the Laplace posterior normal approximation corresponding to the prior $\beta_0, \dots, \beta_m \overset{\text{i.i.d}}{\sim} \text{uniform}(-\infty, \infty)$. Compute the posterior standard errors for the coefficients. Check the correctness of your answers using an inbuilt library function for fitting logistic regression.

4. Consider the linear regression model:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

for $i = 1, \dots, n$ with $\epsilon_i \overset{\text{i.i.d}}{\sim} \text{Lap}(0, \sigma)$. Recall that $\text{Lap}(0, \sigma)$ is the Laplace or double exponential density:

$$x \mapsto \frac{1}{2\sigma} \exp\left(-\frac{|x|}{\sigma}\right)$$

Consider the usual prior:

$$\beta_0, \beta_1, \log\sigma \overset{\text{i.i.d}}{\sim} \text{Unif}(-\infty, \infty).$$

The goal is to fit this model to $(x_1, y_1), \dots, (x_n, y_n)$.

a) Prove that

$$f_{\beta_0, \beta_1 | \text{data}}(\beta_0, \beta_1) \propto \left(\frac{1}{S(\beta_0, \beta_1)}\right)^n \tag{5}$$

where

$$S(\beta_0, \beta_1) := \sum_{i=1}^{n} |y_i - \beta_0 - \beta_1 x_i|.$$

b) Consider the Pearson father-son height dataset that we used in class in Lecture 16. Fit this model to the data and report point estimates and standard errors of β_0 and β_1. You can approximate the joint density (5) by the discrete distribution over a dense grid of β_0 and β_1.

c) Compare your estimates of the intercept and slope with the corresponding estimates when fitting the usual regression model with normal errors. Comment on any discrepancies (or similarities) between the two sets of results.